DLPrPPG: Development and Design of Deep Learning Platform for Remote Photoplethysmography

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***Abstract*—This paper presents a comprehensive neural network-based development platform for remote photoplethysmography (rPPG). rPPG is a growing and popular research area, especially with the introduction of deep learning methods that can significantly improve its signal quality and heart rate prediction reliability. However, there are still many problems with the experimental methods in current studies, such as non-standardized and private data, different pre-processing methods, and incomplete or irreproducible experiment methodologies, among others. These problems prevent methods from being compared fairly and lead to lower reliability of the proposed experimental results, hindering progress in this area. For these reasons, we propose an open-source framework to facilitate the design and experimentation of deep learning-based rPPG development, and it's made freely available on GitHub(DLPrPPG). Through our platform we provide ready-to-use implementations of CNN-AE, LSTM, GAN, and Transformer models, whose hyperparameters we can easily and quickly optimize, and efficiently compare in a fair fashion. From our experiments we show that if the parameters of different neural networks are optimized, the performance of older architectures can be on par or even outperform newer ones.**

1. Introduction

As human life expectancy continues to increase, the need for chronic diseases and long-term care is increasing. The measurement of physiological signals in medicine often requires the application of sensors or contact electrodes with conductive gel on the skin, which is not suitable for patients with sensitive or damaged skin [1]. Therefore, non-contact measurement methods are becoming more important. Among them, rPPG is a promising technology. After Verkryusse et al. [2] demonstrated that heart rate could be accurately measured by using a consumer-grade camera and under normal ambient light conditions, rPPG became more and more popular.

However, in practical applications, rPPG often suffers from noise interference caused by illumination or motion [3], which degrade the rPPG signal quality. Therefore, previous studies tried to solve this problem by utilizing traditional signal processing techniques [4] [5]. Although these methods have improved the quality of the acquired signal, there are still many restrictions on the usage and many assumptions need to be met, which are unlikely to be achieved in the actual environment [6]. Thereafter since these traditional signal processing methods were not sufficient for improving the quality of rPPG signals and the accuracy of heartbeat prediction, recent studies have focused on adapting deep learning (DL) techniques.

Although many studies utilizing DL in the field of rPPG claim to have obtained improvements over previous methods, we have encountered some problems and inconsistencies, which can be summarized as follows: (1) using different signal pre-processing procedures, (2) using different rPPG signal processing algorithms, (3) using different datasets for experiments, and (4) some of the proposed methods are unreproducible. These reasons can lead to unfairness in subsequent comparisons and insufficiently rigorous conclusions, which makes the results deceptive.

For the above reasons, we propose DLPrPPG (Deep Learning Design Platform for rPPG), which is designed to convert signals from rPPG to photoplethysmography(PPG), and supports various DL neural network architectures and loss functions. It can systematically evaluate and analyze the same dataset, which will enable researchers to select the most appropriate research methodologies in a more objective way and significantly reduce the entry barriers for newcomers into this field.

2. Related Work

In the past few years, more and more DL methodologies have been adapted to rPPG. One of the advantages of neural networks is that they can obtain good results without the need for the designer to analyze the problem in depth [7]. These approaches can be roughly divided into two categories. The first aims to extract HR values or pulse signals directly from video through an end-to-end method [8], [9], [10]. Meanwhile, the second one aims to generate clean pulse PPG waveforms, by converting the original rPPG signal obtained through traditional rPPG algorithms by passing it through a neural network [6], [11], [12], [13]. For medical diagnosis, the characteristics of the observed signal are more important than the specific HR values [6], and it's easier to evaluate the strengths and weaknesses of the system by the quality of the generated signals [7] [12]. Besides, it's more difficult to assess which part causes the

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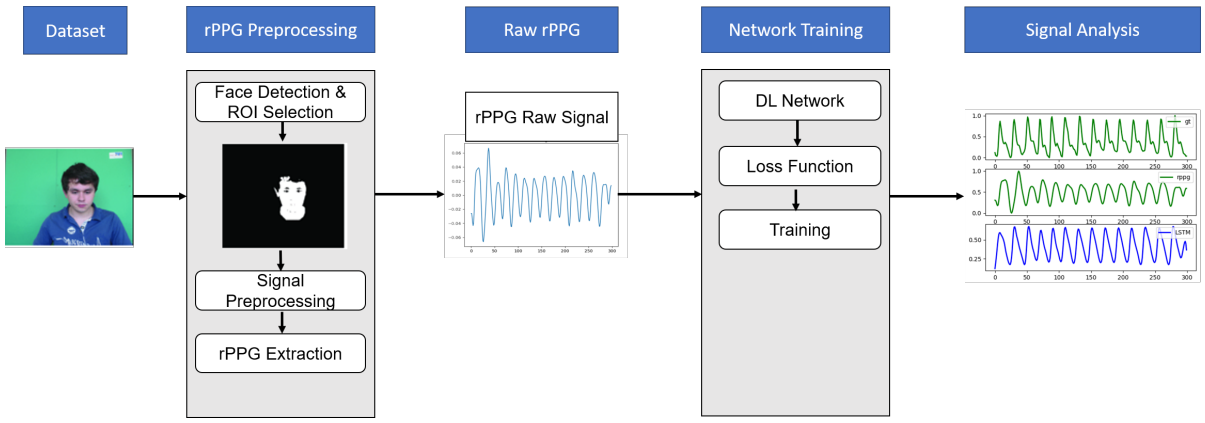


Figure 1. Overview of our framework.

inaccuracy if you combine all functions (e.g. face detection, ROI selection, signal preprocessing, etc.) [6] [11] [12] into an end-to-end black box. By focusing on a specific part we can simplify the problem in order to make improvements. For these reasons, we consider the second method would be better than the first one and therefore, the focus of the neural networks on our framework is to reconstruct PPG signals from rPPG.

3. Methodology

The overall pipeline of our framework is shown in Fig. 1. It can be divided into a traditional rPPG preprocessing and a network training and analysis component. In the first part, we use some signal processing methods to generate the raw rPPG signals. These are then sent into different networks to get the enhanced signal. The rPPG preprocessing part is based on work by Rios et al. [14], which compared several components of a traditional rPPG pipeline including face detection, ROI selection, rPPG algorithm and sliding window size to find out the best combination of those parameters under the consideration of accuracy and computation cost. In this study, we fix those parameters to get the same signals for a fair baseline and focus on training different deep neural network architectures along with the corresponding hyperparameter tuning in order to generate high quality signals. The neural networks provided in this framework and the supported hyperparameter parameters for tuning are shown in Table 1.

1) CNN based Auto-Encoder(CNN-AE): The Auto-Encoder has an encoder stage composed of several 1D convolution layers with strided convolutions for compressed projection and a decoder stage composed of several deconvolution layers, which is designed Symmetrically. Each layer was followed by a PReLUs [15] activation function, except for the output layer where tanh activation was used as a substitute. The number of layers can be adjusted by the parameters according to the design requirements, so it's flexible for users.

2) Long-short term memory (LSTM): We implement the stacked LSTM in our platform. It has several layers of the LSTM hidden layer and can be changed to a bideractional

TABLE 1. SUMMARY OF MODELS AND HYPERPARAMETERS TUNABLE THROUGH OUR FRAMEWORK.

Model	Hyperparameter
CNN-AE	Kernel width (k), number of filters for first layer (n), number of layers (L), batch size (B), β for LE, learning rate (lr)
LSTM	Hidden size (H), number of layers (L), bidirectional, batch size (B), β for LE, learning rate (lr)
GAN	Kernel width (k), number of filters for first layer (n), batch size (B), learning rate (lr)
	Training style(Loss function type): cGAN/WGAN
	Improvement method: WGAN-GP/SN-GAN
Transformer	Kernel width(k), number of layers (L), embed size (E), heads (head), batch size(B), β for LE, learning rate

LSTM(BiLSTM) [16] for enhanced signal quality. Then the enhanced output is passed by a fully connected layer (also known as dense layer) and a tanh activation.

3) Generative Adversarial Network (GAN) [17]: The structure of our GAN is based on PulseGAN [6], which is a type of cGAN [18]. To improve the training stability, we adapt two methods to generate much better and stable result. The WGAN-GP [19] changed the loss function of the original GAN, while the SN-GAN [20] introduce a normalization method called spectral normalization. Their main goal is to limit the discriminator to fit Lipschitz constraint [19] and guide the generator to generate the data which has the same distribution of the real signal.

4) Transformer [21]: The main structure of the transformer in our platform is basically from the encoder stage of Informer [22], however for the transformer block we change it to the vanilla transformer encoder and we don't use convolutions for downsampling the data (as in Informer), which is similar to the structure of the LSTM in our platform.

For the loss function, we implement two simple and classic loss functions, which are mean absolute error (L1) loss and mean squared error (L2) loss; we compute the prediction error between generated signal and ground truth PPG. Also, we implement a loss function which we refer to as L1 loss-expand (LE), which is adapted from the L_{QRS}

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loss proposed by Chiu et al. [23], but for usage with PPG signals.

We evaluate our DLPrPPG framework on UBFC dataset [24], which has 42 videos synchronized with PPG ground truth. The videos are about 1 to 2 minutes utilizing a Logitech C920 HD Pro webcam, at 30 FPS with a resolution of 640x480 in uncompressed 8-bit RGB format. For the parameter in our signal pre-processing component we choose $skin$ for ROI selection, which takes the whole skin region in the face into consideration, $deep$ for face detection, which is deep learning-based SSD (single-shot detector) with ResNet-based architecture, and the traditional rPPG algorithm POS [25], as described in [14]. We choose 10-second sliding window (SW) size and sliding window step is taken as 0.5 seconds to increase the number of samples, so there are $SW \times FPS$ data points for every samples. Since the actual frame rate of each video is not the same (it's around 20 to 30fps) and that would cause the number of data points between samples to be different, we first resample each sample to 30 FPS. There were 2641 samples in the training set, 661 samples in the validation set, and 1385 samples in the testing set. Since the scale of the ground truth PPG and obtained rPPG signals is not the same, these signals were first min-max scalar normalized to [0, 1] for training.

All experiments were implemented in PyTorch, and conducted on a workstation equipped with Intel Xeon Gold 6248 CPU, and a NVIDIA Quadro RTX 8000 GPU. We utilize the Adam optimizer, and all of the neural networks were trained for 50 epochs. We used the "ReduceLROnPlateau" learning rate scheduler with the factor to 0.5 and patience to 3.0 to reduce the learning rate dynamically except for the GAN models because the behavior of the loss of GANs is different from other neural networks, and in our experiments it fails to converge.

We define different metrics to evaluate the quality of the signals generated by different networks. First, we define the average HR of each sample, which could be estimated by the average interval beat interval (IBI) [26]. Compared with the previous methods which transferred the signals into frequency domain to estimate the HR, it could reflect the characteristics of the signal more accurately. The $\overline{IBI}$ was the average of the IBI in the sliding window:

$$HR = 60/\overline{IBI}, \tag{1}$$

Then we compute the Mean Absolute Error(MAE), the Root-Mean-Square Error(RMSE) and the Pearson Correlation Coefficient(PCC), between the predicted rPPG signal and HR, and its corresponding ground truth PPG and HR.

4. Results and Discussion

4.1. Model Parametric Study

Fig. 2 shows the experiment results of the CNN-AE using different kernel widths. From this figure we can observe that the performance would be better if the kernel width is less than 7, and the training would fail when the kernel width is larger than 9. This is in contrast to other studies which usually set kernel width to 31 or didn't include this information. Since the characteristics of each data distribution are different, and different pre-processing methods are used, the most suitable design hyperparameters should be evaluated on a case by case basis. We should optimize different models for different use cases and not just blindly adopt the original setup from other studies.

Fig. 3 shows the results for different loss function for the CNN-AE, LSTM and the Transformer. We compare L2, L1 and different β values for LE loss. The best result is obtained when β is between 0 to 0.7, while it gets worse as β value gets larger than 1. We suggest to try these three

TABLE 2. SUMMARY OF BEST RESULTS OBTAINED FOR EACH METHOD.

Method	Signal			HR [BPM]		
	MAE	RMSE	PCC	MAE	RMSE	PCC
POS	0.293	0.293	0.393	3.310	5.810	0.848
CNN-AE	**0.187**	**0.187**	**0.461**	2.524	**4.366**	**0.909**
LSTM	0.187	0.187	0.479	2.625	4.544	0.900
GAN	0.245	0.245	0.378	2.567	4.473	0.904
Transformer	0.193	0.193	0.432	**2.505**	4.643	0.897

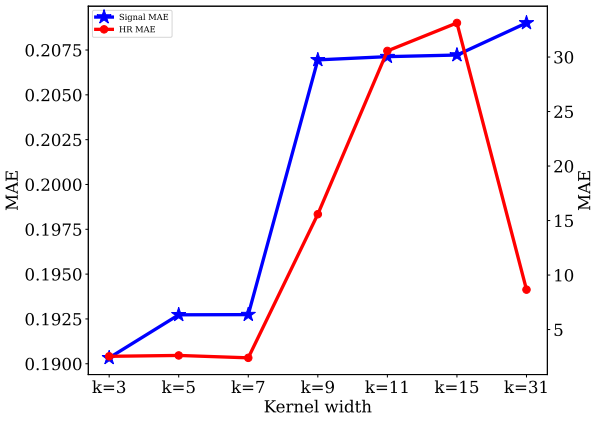


Figure 2. Different kernel width k for CNN-AE. The blue line is the signal MAE and the red line is the HR MAE.

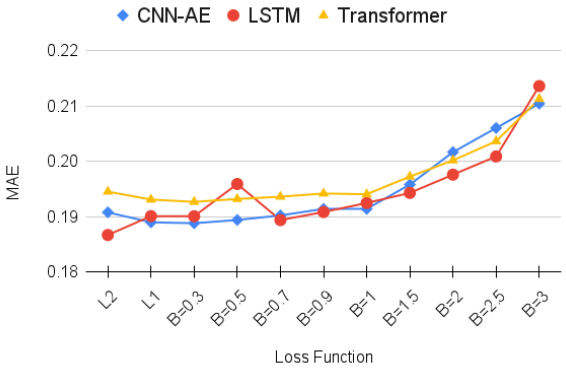
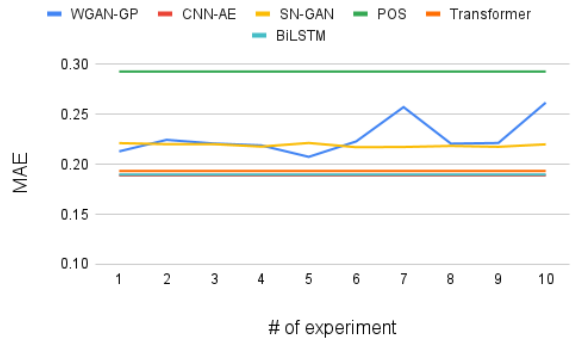
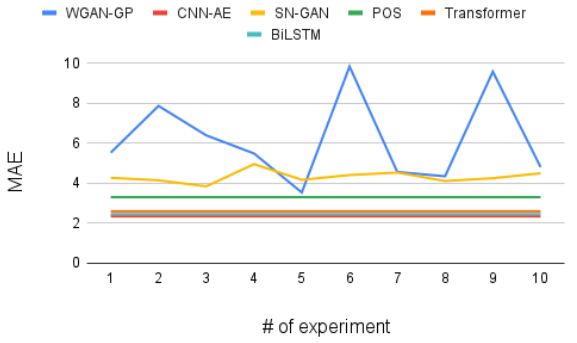


Figure 3. Results of signal MAE for different loss functions.

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(a)



(b)

Figure 4. Results consistency of different models. (a) is the signal MAE; (b) is the HR MAE.

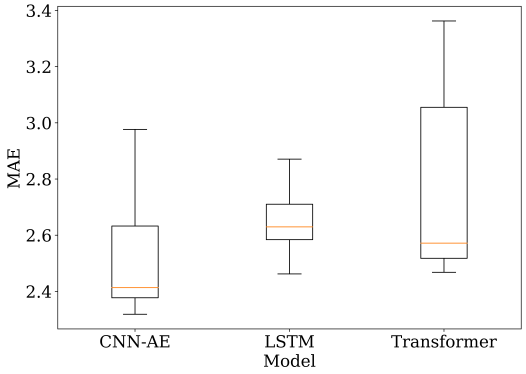


Figure 5. Box plot for all different runs across different models.

methods when fine-tuning different models to find out the best results.

We test the consistency of different models. We repeat each experiment ten times for different models and different combination of hyperparameters. Unlike other models, there is great instability in the training process of GAN. First, we utilize WGAN-GP and SN-GAN to solve the gradient vanishing problem that would happen with the original PulseGAN and its loss function. However, using these two methods would still lead to some inconsistency problems due to the unique loss landscape of GANs, which aim to reach a saddle point, rather than a local minima as in other models [27]. This can be seen from Fig. 4, which shows that the experimental results for the same set of hyperparameters may be wildly different, even with a fixed random seed, in contrast to other models which are consistent. Due to this we advise against using GANs in this application unless it is necessary, otherwise, finetuning would take a long time.

4.2. Model Comparison

Table. 2 summarize the best result for different models from all of our experiments. The result of GAN was obtained by SN-GAN. We found that the CNN-AE outperforms other networks, which is different from the conclusion of previous studies; many experiments use the CNN-AE as a baseline, and then claim that the new model can improve upon it. This happens because in their study the CNN-AE design hyperparameters are fixed while they tune hyperparameters for their proposed model. From this we can conclude that older neural network architectures do not necessarily perform worse if hyperparameters are carefully tuned for different experiments, a conclusion similar to the one reached by Lucic et al. [28]. Therefore, we suggest to fairly compare every model to avoid reaching misleading conclusions.

Fig. 5 shows the box plot for HR MAE of different models across all their different hyperparameter configurations. We don't compare GAN because we've showed its instability in Fig. 4. The LSTM has the smallest interquartile range (IQR), which represents that in general, its more robust to hyperparameter settings, while the Transformer has the largest, representing it's more sensitive to hyperparameter selection. For CNN-AE, the median is lower than the other two models, so it's easy to get good results when doing hyperparameter tuning.

5. Conclusion

We propose an open-source framework to facilitate the design and experimentation of deep learning-based neural network development for rPPG. Through our platform we can avoid common pitfalls found in the novel method design and experimentation process, which often lead to incomplete results and misleading conclusions. With our platform, we can quickly select the optimal parameters for different neural networks. For example, in the CNN-AE, by optimizing the kernel width, the resulting signal MAE is reduced from 0.209 to 0.190, while the HR MAE is reduced from 33.127 to 2.405. These factors are rarely considered in other studies, or only a few specific items are compared, but with our platform, the influence of different combination of parameters can be easily highlighted. Additionally, through our platform, we have shown that if the parameters of different neural networks are optimized, the performance of older methods is not necessarily worse than that of the newer ones.

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